

**VISVESVARAYA TECHNOLOGICAL
UNIVERSITY BELAGAVI-590 018, KARNATAKA.**



TECHNICAL SEMINAR REPORT

ON

“PROMPT ENGINEERING IN HEALTHCARE”

Submitted in the partial fulfillment of requirements for the award of Degree

B.E. in Computer Science and Engineering

SEMINAR ASSOCIATE

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4BD21CS020

SEMINAR GUIDE

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2024-2025

**Bapuji Institute of Engineering and Technology
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CERTIFICATE

This is to certify that **ANUSHREE B KUPPAST** bearing USN **4BD21CS020** of Computer Science and Engineering department has satisfactorily submitted the Technical Seminar Report entitled “**PROMPT ENGINEERING IN HEALTHCARE**” for **TECHNICAL SEMINAR** in partial fulfillment of the requirements for the award of Degree of Bachelor of Engineering (B.E.) in Computer Science & Engineering, under the VTU during the academic year 2024-25.

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Bapuji Educational Association (Regd.)
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Vision and Mission of the Institute

Vision

—To be a center of excellence recognized nationally internationally, in distinctive areas of engineering education and research, based on a culture of innovation and invention.¶

Mission

—BIET contributes to the growth and development of its students by imparting a broad-based engineering education and empowering them to be successful in their chosen field by inculcating in them positive approach, leadership qualities and ethical values.¶

Vision and Mission of the Computer Science and Engineering Department

Vision

—To be a center-of-excellence by imbibing state-of-the-art technology in the field of Computer Science and Engineering, thereby enabling students to excel professionally and be ethical.¶

Mission

1.	Adapting best teaching and learning techniques that cultivates Questioning and Reasoning culture among the students.
2.	Creating collaborative learning environment that ignites the critical thinking in students and leading to the innovation.
3.	Establishing Industry Institute relationship to bridge skill gap and make them industry ready and relevant.
4.	Mentoring students to be socially responsible by inculcating ethical and moral values.

Program Educational Objectives (PEOs):

PEO1	To apply skills acquired in the discipline of computer science and engineering for solving Societal and industrial problems with apt technology intervention.
PEO2	To continue their carrier ion industry /academia or pursue higher studies and research.
PEO3	To become successful entrepreneurs, innovators to design and develop software products and services that meets societal, technical and business challenges.
PEO4	To work in the diversified environment by acquiring leadership qualities with effective communication skills accompanied by professional and ethical values.

Program Specific Outcomes (PSOs):

PSO1	Analyze and develop solutions for problems that are complex in nature but applying the knowledge acquired from the core subjects of this program.
PSO2	To develop secure, scalable, resilient and distributed applications for industry and societal Requirements.
PSO3	To learn and apply the concepts and contract of emerging technologies like artificial intelligence, machine learning, deep learning, big-data analytics, IOT, cloud computing etc. for any real time problems.

ABSTRACT

Prompt engineering is a powerful technique in AI that helps improve how models understand and respond to user inputs. This project, "AI-Powered Symptom Checker," applies prompt engineering in healthcare to generate possible medical conditions based on symptoms entered by users. It uses Google's FLAN-T5-small model, a transformer-based AI, to process symptoms and provide meaningful responses. The system is built using Streamlit for an easy-to-use interface and Hugging Face's transformers library for text generation. Well-structured prompts guide the model to give accurate and relevant medical insights, making the AI responses more reliable. The project also includes Torch and asyncio optimizations for smooth execution. This project highlights how carefully designed prompts can improve AI's role in healthcare, helping users get better symptom-based guidance. Future enhancements may include fine-tuning the AI with medical data and integrating reliable medical databases for more precise results.

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